

Traffic Light Control System: FreeRTOS (WIN32-MSVC)

Overview

This document is a print-ready report for the FreeRTOS-based traffic light control system running on the official Windows (MSVC/MingW) simulator. The system coordinates North–South (NS) and East–West (EW) traffic lights, synchronizes pedestrian signals, handles emergency vehicle preemption, and logs every state change with timestamps to `traffic_log.txt`. The Visual Studio console provides a live operational trace.

1. Architecture

- Tasks: `ControllerTask` sequences NS/EW phases and pedestrian timing; `EmergencyTask` simulates emergency detection and asserts an Event Group bit for immediate preemption.
- Synchronization: An Event Group instantly interrupts delay windows; a binary semaphore releases the controller when the emergency clears.
- Logging: State changes mirror to console and write to `traffic_log.txt` with timestamps.
- Memory allocator: `heap_5.c` is used. Memory regions are defined with `vPortDefineHeapRegions()` via `prvInitialiseHeap()` before creating RTOS objects.

2. Timing Specification

Direction	State	Duration
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North–South	Green	10 s
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North–South	Yellow	3 s
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North–South	Red	11 s (EW Green 8 s + Yellow 3 s)
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East–West	Green	8 s
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East–West	Yellow	3 s
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East–West	Red	13 s (NS Green 10 s + Yellow 3 s)
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3. Pedestrian Synchronization

- NS pedestrians: WALK for 8 s at start of NS RED, then DON'T WALK for 3 s.
- EW pedestrians: WALK for 6 s at start of EW RED, then DON'T WALK for 7 s.

4. Emergency Vehicle Preemption

EmergencyTask preempts the normal phase cycle instantly, forcing a safe-priority state (EW=GREEN, NS=RED, all pedestrians DON'T WALK). After the 5 s emergency window, control returns to ControllerTask via semaphore release.

5. Print Preview with Explanations

NS=GREEN, EW=RED, EW Ped=WALK at start of EW RED

This phase shows the North–South direction receiving right-of-way while East–West remains RED. Because EW is RED, the EW pedestrian signal enters its WALK window. NS pedestrians remain DON'T WALK.

```
[1773592310] North-South Light: GREEN
[1773592310] East-West Light: RED
[1773592310] East-West Pedestrian: WALK
[1773592310] NS:GREEN  EW:RED  |  NS Ped:DON'T WALK  EW Ped:WALK
```

NS transitions to YELLOW (3 s)

This captures the transition from NS GREEN to NS YELLOW. The 3-second yellow warns drivers of the upcoming stop. Both pedestrian signals remain DON'T WALK for safety during this transition.

```
[1773592320] North-South Light: YELLOW
[1773592320] NS:YELLOW  EW:RED  |  NS Ped:DON'T WALK  EW Ped:DON'T WALK
```

EW=GREEN, NS=RED, NS Ped=WALK

This is the mirror of Screenshot 1. Here EW gets GREEN and NS turns RED. NS pedestrians WALK during the first 8 seconds of the NS RED window. EW pedestrians remain DON'T WALK.

```
[1773592323] North-South Pedestrian: WALK
[1773592323] East-West Light: GREEN
[1773592323] NS:RED  EW:GREEN  |  NS Ped:WALK  EW Ped:DON'T WALK
```

*** EMERGENCY DETECTED! PRIORITIZING TRAFFIC ***

Emergency detection forces an immediate override of normal sequencing. NS goes RED, EW goes GREEN, and all pedestrians are forced to DON'T WALK to clear the intersection safely.

```
[1773592341] *** EMERGENCY DETECTED! PRIORITIZING TRAFFIC ***  
[1773592341] North-South Light: RED  
[1773592341] East-West Light: GREEN  
[1773592341] North-South Pedestrian: DON'T WALK  
[1773592341] East-West Pedestrian: DON'T WALK  
[1773592341] NS:RED EW:GREEN | NS Ped:DON'T WALK EW Ped:DON'T WALK
```

6. Log Excerpt (traffic_log.txt)

```
[1773592310] North-South Light: RED  
[1773592310] East-West Light: RED  
[1773592310] North-South Pedestrian: DON'T WALK  
[1773592310] East-West Pedestrian: DON'T WALK  
[1773592310] NS:RED EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK  
[1773592310] North-South Light: GREEN  
[1773592310] East-West Light: RED  
[1773592310] East-West Pedestrian: WALK  
[1773592310] NS:GREEN EW:RED | NS Ped:DON'T WALK EW Ped:WALK  
[1773592316] East-West Pedestrian: DON'T WALK  
[1773592316] NS:GREEN EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK  
[1773592320] North-South Light: YELLOW  
[1773592320] NS:YELLOW EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK  
[1773592323] North-South Light: RED  
[1773592323] NS:RED EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK  
[1773592323] North-South Pedestrian: WALK  
[1773592323] East-West Light: GREEN  
[1773592323] NS:RED EW:GREEN | NS Ped:WALK EW Ped:DON'T WALK  
[1773592331] North-South Pedestrian: DON'T WALK
```

[1773592331] NS:RED EW:GREEN | NS Ped:DON'T WALK EW Ped:DON'T WALK
 [1773592331] East-West Light: YELLOW
 [1773592331] NS:RED EW:YELLOW | NS Ped:DON'T WALK EW Ped:DON'T WALK
 [1773592334] East-West Light: RED
 [1773592334] NS:RED EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK
 [1773592334] North-South Light: GREEN
 [1773592334] East-West Light: RED
 [1773592334] East-West Pedestrian: WALK
 [1773592334] NS:GREEN EW:RED | NS Ped:DON'T WALK EW Ped:WALK
 [1773592341] *** EMERGENCY DETECTED! PRIORITIZING TRAFFIC ***
 [1773592341] North-South Light: RED
 [1773592341] East-West Light: GREEN
 [1773592341] North-South Pedestrian: DON'T WALK
 [1773592341] East-West Pedestrian: DON'T WALK
 [1773592341] NS:RED EW:GREEN | NS Ped:DON'T WALK EW Ped:DON'T WALK
 [1773592346] *** EMERGENCY CLEARED ΓÇö returning to normal cycle ***
 [1773592346] North-South Light: GREEN
 [1773592346] East-West Light: RED
 [1773592346] East-West Pedestrian: WALK
 [1773592346] NS:GREEN EW:RED | NS Ped:DON'T WALK EW Ped:WALK
 [1773592352] East-West Pedestrian: DON'T WALK
 [1773592352] NS:GREEN EW:RED | NS Ped:DON'T WALK EW Ped:DON'T WALK
 [1773592356] North-South Light: YELLOW

7. Build & Run Notes (WIN32–MSVC)

- Use the official WIN32–MSVC FreeRTOS Windows simulator.
- Only one heap scheme should be compiled (heap_5.c for this project).
- Ensure prvInitialiseHeap() is called before task creation.
- Logs are written to traffic_log.txt automatically during simulation.